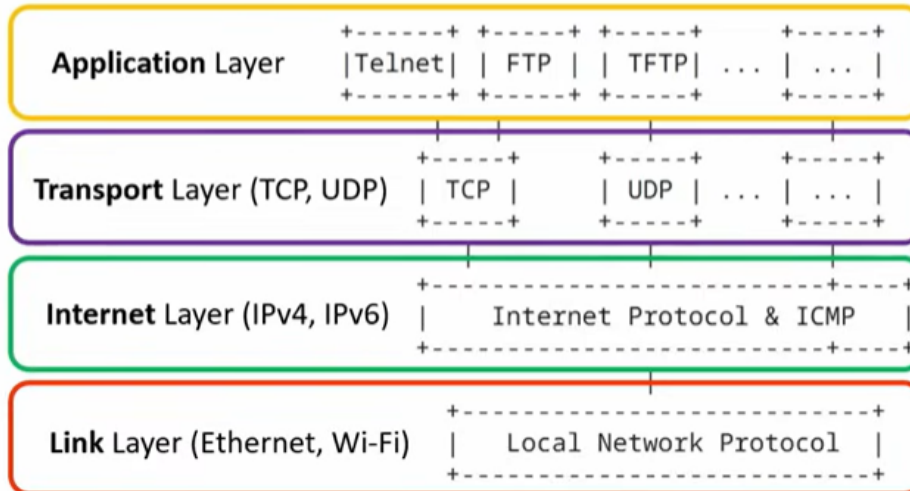




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Layered Models

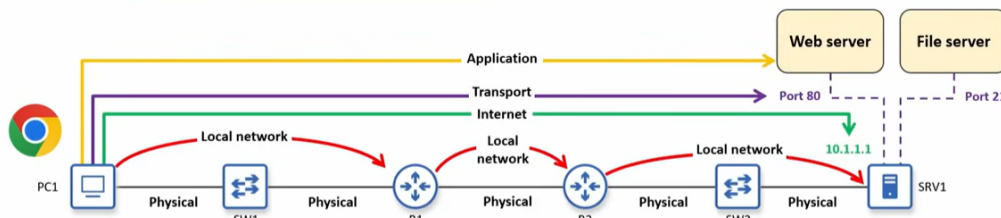


From RFC 791, "Internet Protocol"
(1981)



TCP/IP Model

Layer	Name	What It Does	Examples
5	Application	Provides network services to applications	HTTP/HTTPS, DNS, DHCP, FTP, SMTP, SSH
4	Transport	Provides end-to-end communication and uses port numbers	TCP, UDP, ports 80, 443, 53
3	Internet	Handles logical addressing and routing between networks	IPv4, IPv6, ICMP, routers
2	Local Network Layer	Handles local network delivery using MAC addresses and frames	Ethernet, Wi-Fi, switches, MAC addresses
1	Physical	Transmits raw bits across the physical medium	Copper cables, fiber optic, radio/Wi-Fi signals, connectors



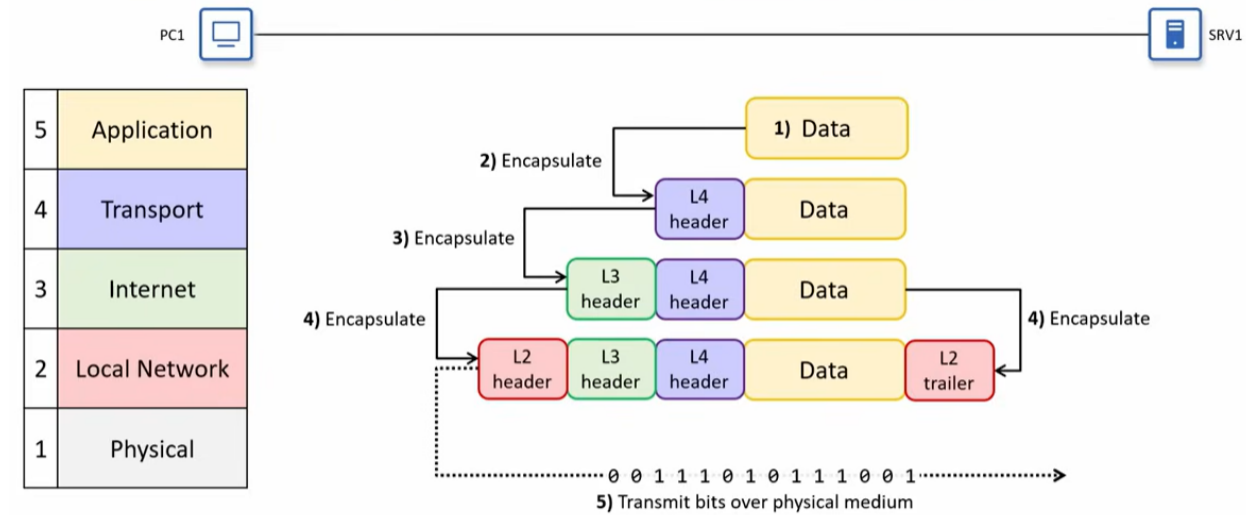
Application layer	• Protocols for communication between application processes; create and interpret the data.
Transport layer	• Provides end-to-end communication between application processes using port numbers .
Internet layer	• Provides end-to-end communication between hosts across networks using IP addresses and routers.
Local Network layer	• Provides hop-to-hop delivery within a local network using MAC addresses and switches.

Physical layer

- Sends bits as electrical, optical, or radio signals over the physical medium.



Encapsulation and decapsulation



Stage / Layer	What Gets Added	Name of Data	PDU Name	Protocol Example
Layer 7–5	Application data	Data (payload)	L7–L5 PDU	HTTP, DNS, etc.
Layer 4 – Transport	L4 header	Segment (TCP) or Datagram (UDP)	L4 PDU	TCP / UDP
Layer 3 – Network	L3 header	Packet	L3 PDU	IP
Layer 2 – Data Link	L2 header + trailer	Frame	L2 PDU	Ethernet
Layer 1 – Physical	Converted to electrical/light/radio signals	Bits	L1 PDU	Ethernet cable, Wi-Fi

A PDU is a Protocol Data Unit. Each step of the process is a PDU.

OSI MODEL

OSI model

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

